SYSTEM WALKTHROUGH

From Bibliometric Data to Transferable Scientific Knowledge



An evidence-grounded paper attribution system for cross-SDL discovery

Classification · Attribution · Review · Knowledge Transfer

UNIVERSITY OF TORONTO · 2026

The design principle: every label must remain reusable

A bibliometric dataset becomes useful when each paper can be traced from raw evidence to downstream knowledge.

01

NORMALIZED

One canonical paper, author, institution, and source identity.

02

CLASSIFIED

A label with explicit evidence signals and necessary conditions.

03

ATTRIBUTED

A defensible relationship to AC, SDL practice, or adjacent fields.

04

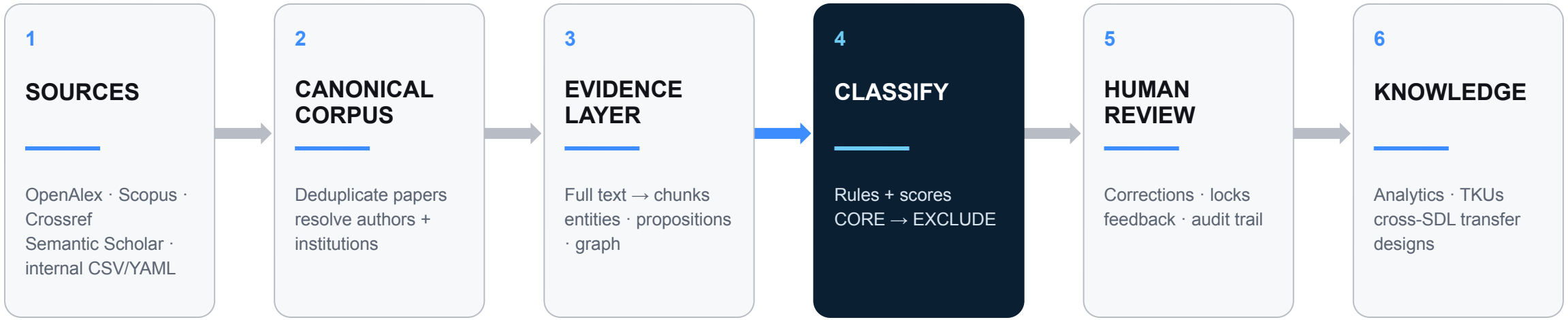
AUDITABLE

The exact snippets, paths, rules, and human decisions remain inspectable.

Not a generic paper manager. Not a chatbot that invents facts. A controlled transformation from sources to decisions.

The system in one view

Six controlled transformations turn heterogeneous publication data into reviewable knowledge assets.



HARD BOUNDARY

LLMs may interpret approved evidence. They may not invent papers, graph edges, measurements, citations, or numerical claims.

Evidence starts before classification

Identity resolution and full-text provenance determine how trustworthy every later algorithm can be.

HETEROGENEOUS INPUTS

METADATA	DOI · title · abstract · year · venue
IDENTITY	authors · ORCID · affiliations · aliases
FULL TEXT	PDF/HTML → parsed text with provenance
CURATION	AC members · partners · taxonomy · gold papers

NORMALIZATION OPERATORS

01 Deduplicate papers

DOI, normalized title, source reconciliation

02 Resolve people + institutions

exact, aliases, initials, contextual fuzzy match

03 Segment evidence

sections, chunks, entities, propositions, citations

CANONICAL EVIDENCE RECORD

PAPER

Stable identity + source provenance

- linked authors and institutional identities
- topic and SDL evidence candidates
- source-addressable text chunks
- citations, entities, propositions, graph edges
- review state and decision history

Classification is a hybrid evidence system

The production decision combines deterministic evidence families; no single model is allowed to define relevance.

A	**AUTHOR**	exact · alias · initials · contextual fuzzy
I	**INSTITUTION**	partner aliases · affiliation normalization
T	**TOPIC**	taxonomy phrases · negation · density · embedding
F	**FULL TEXT**	funding · acknowledgements · methods · captions
N / Q	**GRAPH**	author proximity · excess centrality · citations
E	**ENTITIES**	methods · instruments/software · materials

DECISION LOGIC

Signals answer different questions.

01 Who is involved?

02 Where is the work anchored?

03 What scientific problem is addressed?

04 How is the work performed?

05 How close is it to known AC research?

Result: a label plus inspectable evidence — not a probability alone.

Lexical precision first, embeddings second

The topic scorer protects scientific meaning with phrase boundaries, negation checks, blacklists, and density before optional semantic similarity.



KEYWORD EVIDENCE

$$K = \min\left(\Sigma \text{ kept phrase weights} / 2 , 1 \right)$$

OPTIONAL SEMANTIC BLEND

$$T = (1 - \lambda)K + \lambda \cdot \text{cosine}$$

SCIENTIFIC SAFEGUARDS

NEGATION	“not a closed loop” does not become positive SDL evidence	BLACKLIST	high-risk generic phrases can veto a false positive
DENSITY	≥2 distinct topic groups adds a small corroboration boost	ADJACENT	adjacent-only evidence is damped and capped below core

$\lambda = 0.4$ by default

Embeddings enrich a guarded lexical score; they do not replace it.

Section-aware, weighted, and capped

Evidence is extracted from semantically different regions of the paper, then aggregated without allowing one repeated phrase to dominate.

EVIDENCE TYPE	TYPICAL LOCATION	WEIGHT PER HIT	TYPE CAP
Funding acknowledgement	Acknowledgements / funding	**+0.60**	**1.00**
Partner institution acknowledgement	Acknowledgements / affiliations	**+0.30**	**0.60**
SDL methodology	Methods / experimental workflow	**+0.20**	**0.70**
SDL figure caption	Figures / captions	**+0.20**	**0.50**
Disqualifying evidence	Any source-addressable section	**−0.40**	**penalty**

FULL-TEXT EVIDENCE SCORE

$$F = \text{clamp}(\Sigma \text{ type-capped evidence weights} / 1.5 , 0 , 1)$$

Why caps matter

Repeated boilerplate cannot overwhelm independent evidence families.

Confidence ranks cases; necessary conditions assign labels

The weighted blend summarizes evidence strength, while a rule tree protects semantic meaning and high-value labels.

WHEN FULL TEXT IS AVAILABLE

$$C = .35A + .15I + .25T + .20F + .10N + .10Q + .05E - .15D$$

METADATA-ONLY FALLBACK

$$C = .40A + .20I + .30T + .10N + .10Q + .05E - .15D$$

A author · I institution · T topic · F full text · N network · Q citations · E entities · D disqualifiers

Professional takeaway

A high confidence score cannot bypass the evidence required for a high-value label.

CORE	Gold or direct AC identity + core SDL evidence
RELATED	Strong identity, funding, or AC-core citation support
ADJACENT	Strong topic, full text, second-degree, or entity evidence
UNCERTAIN	Partial evidence inside the review band
EXCLUDE	No necessary condition survives

Scopus uses two independent questions

A paper can be scientifically relevant to SDLs without being directly attributable to the Acceleration Consortium.

AC ATTRIBUTION

AC = .35 acknowledgement + .30 funding + .20 member + .10 high-relevance + .05 retrieval

SDL RELEVANCE

SDL = .45 architecture + .30 retrieval + .25 keyword / tag

ARCHITECTURE COMPLETENESS

AI decision-making + **automation** + **closed loop**

architecture = completed dimensions / 3

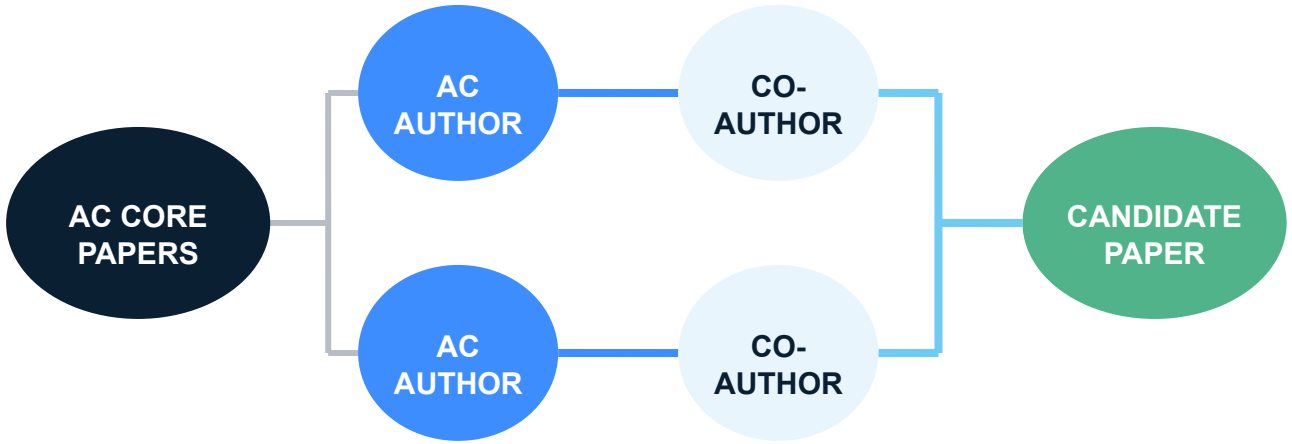
↑ SDL RELEVANCE

HIGH AC HIGH SDL CORE or RELATED	**LOW AC HIGH SDL** SDL FIELD RELEVANT
HIGH AC LOW SDL RELATED	**LOW AC LOW SDL** ADJACENT / UNCERTAIN / EXCLUDE

AC ATTRIBUTION →

Graph structure extends attribution without a production GNN

The current graph signal is deterministic: personalized PageRank, excess centrality, citation proximity, and stored second-degree paths.



Personalized flow from known AC evidence

second-degree paths are retained for inspection

AUTHOR NETWORK SCORE

$$N(p) = \max_a PPRAC(a) \times [\, 1 + .25 \cdot \max(0, cnorm(a) - 1) \,]$$

CITATION SCORE · CAC = UNIQUE AC-CORE CITATIONS

$$Q(p) = \min(\, 1 \,,\, 0.5 \times |CAC(p)| \,)$$

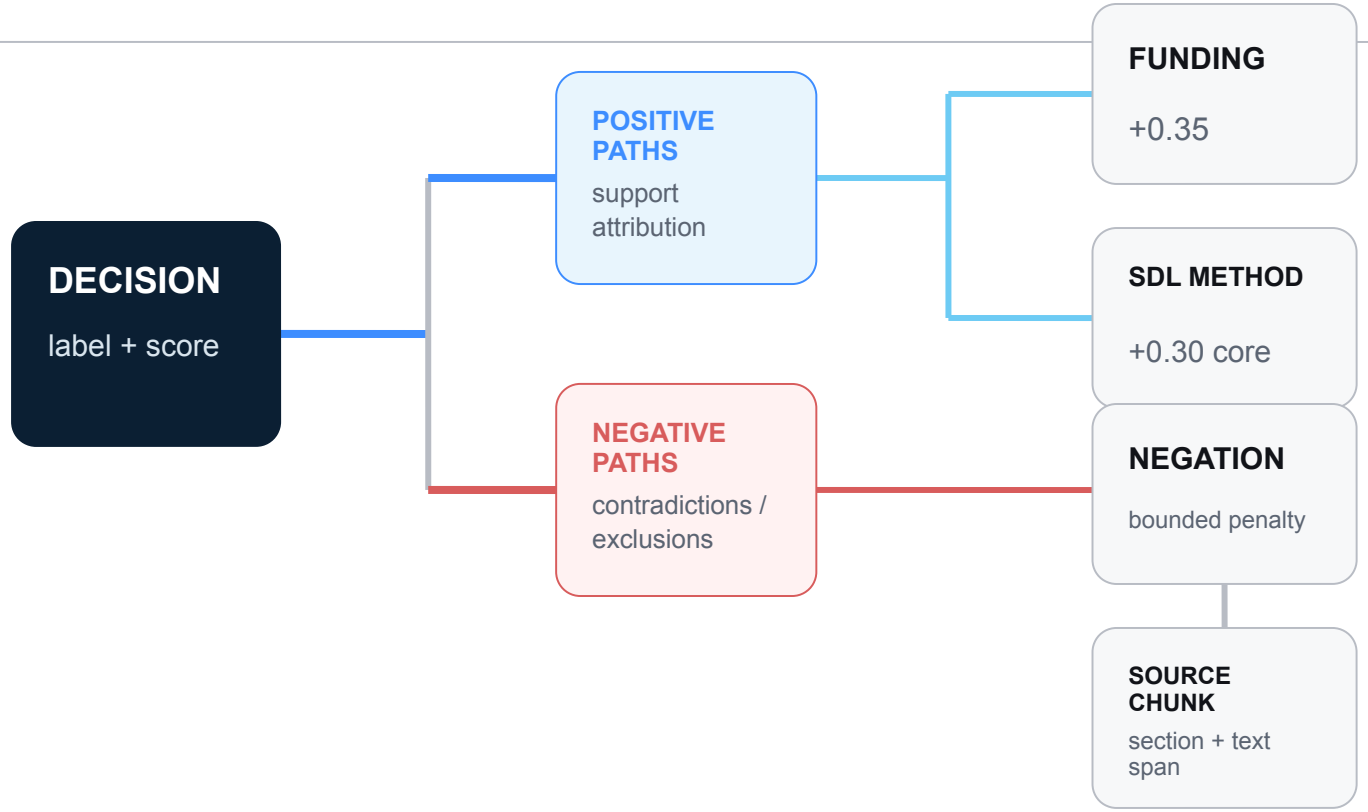
PRODUCTION BOUNDARY

α = 0.85 · weighted normalized coauthor graph

GNN work remains an offline baseline / future experiment.

The AET makes each decision inspectable

Instead of one opaque score, the system can expose positive and negative evidence paths back to source-addressable chunks.



EVIDENCE-TREE SCORE

$$S(AET) = \text{clamp}(\Sigma \text{ positive path weights} - \Sigma \text{ negative path weights} , 0 , 1)$$

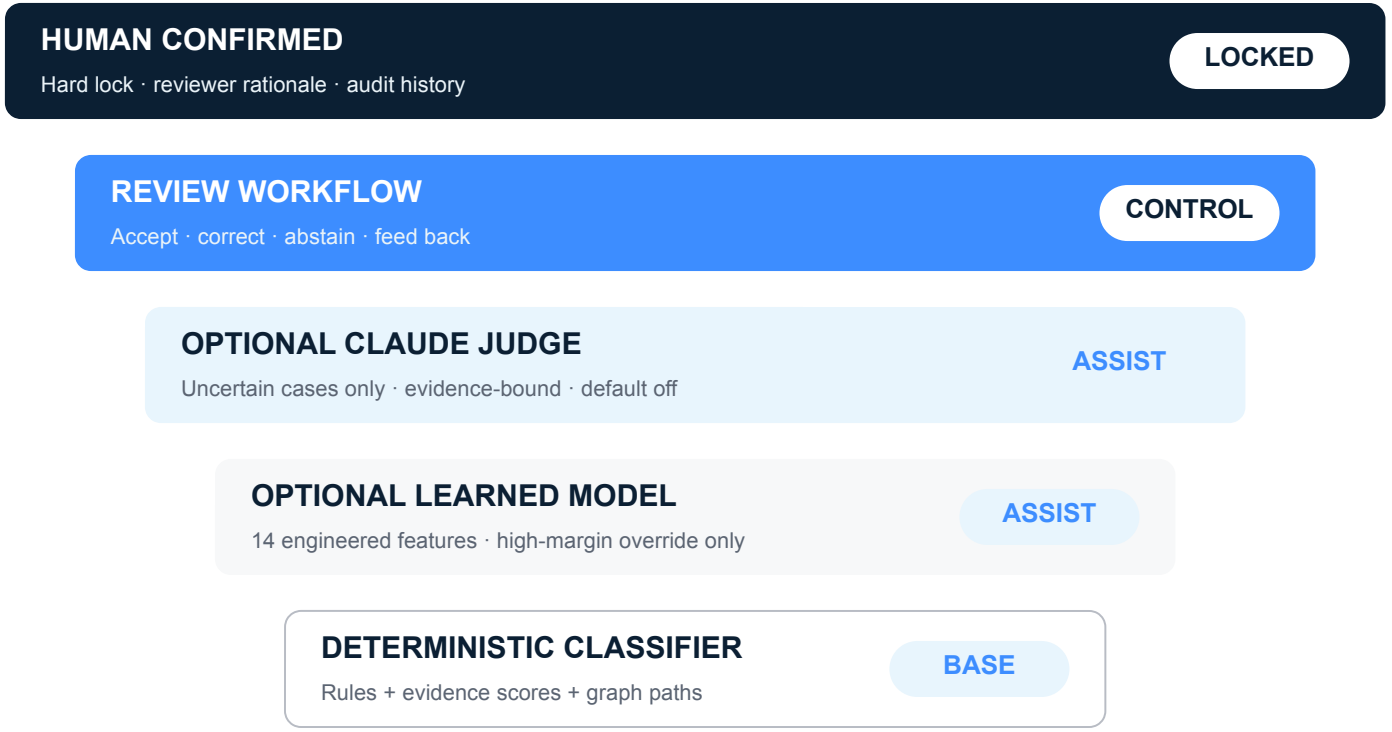
REPRESENTATIVE PATH WEIGHTS

Funding evidence	+0.35
Partner affiliation	+0.30
Core SDL methodology	+0.30
AC author identity	+0.25
Negative evidence	capped at 0.25

A decision can be challenged at the path level — not only at the final label.

Human review controls model authority

Optional models can prioritize uncertain cases and explain evidence, but they remain bounded by deterministic data and human-confirmed locks.



SAFE ROLE FOR CLAUDE

Fund adjudication, not truth generation.

- summarize retrieved evidence
- judge only an uncertainty queue
- draft reviewer rationale
- generate citations only from approved sources

Never invent papers, edges, numbers, results, or citations.

Measure reliability at the decision boundary

A trustworthy bibliometric dataset is evaluated across identity quality, label quality, evidence quality, and operational behavior.

DATASET QUALITY

Duplicate rate
identity-resolution precision
full-text coverage

CLASSIFICATION

per-label precision / recall / F1
macro-F1 + confusion matrix
CORE / RELATED retrieval

EVIDENCE

path precision@k
source-chunk coverage
negative-evidence recall

OPERATIONS

abstention rate
human correction rate
cost + latency per adjudication

RELEASE GATE

Labels, evidence paths, and review behavior must pass together.

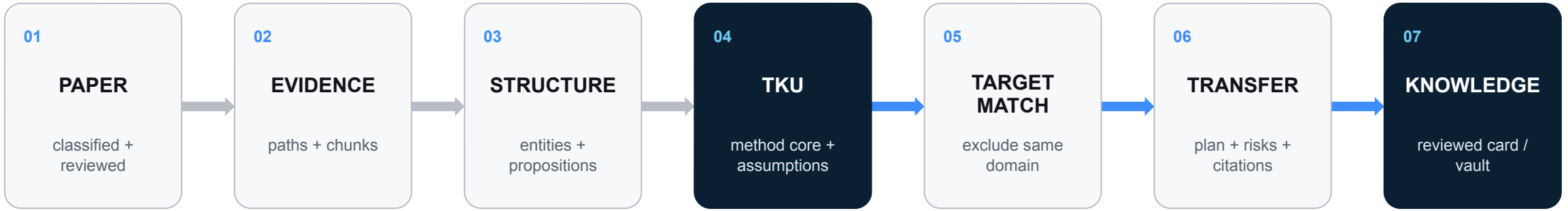
GNN STATUS

Offline structural baseline only

No deployed PyG / DGL training or inference path.

Classification is the front door to cross-SDL knowledge

Once evidence is source-addressable, the system can extract transferable methods without detaching them from provenance.



EVOLUTIONARY TRANSFER CANDIDATE

Score = match strength + novelty + transferability + evidence − risk penalty

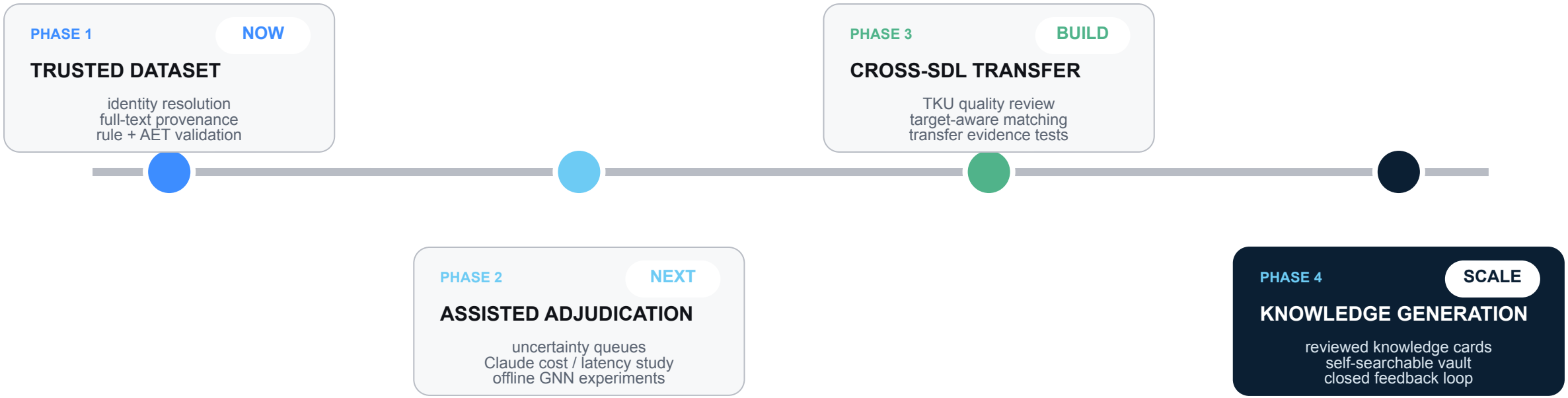
NON-NEGOTIABLE

Every generated claim carries a source tag such as [src:paper_id].

The same domain is excluded during transfer matching; human review precedes knowledge acceptance.

A phased path from dataset to knowledge generation

Each phase adds capability only after the preceding evidence and governance boundary is measurable.



Decision rule: no new model receives production authority until its evidence quality, disagreement behavior, and review burden are measured.

CLOSING SYNTHESIS

Build the dataset so every label can become reusable knowledge.

The system is strongest when classification, evidence, review, and transfer are treated as one provenance-preserving pipeline.

01

DATA SCOPE

Which bibliometric sources and gold sets define the first trusted release?

02

REVIEW POLICY

Which labels and uncertainty bands require human confirmation?

03

MODEL BUDGET

Where should Claude credits accelerate adjudication without expanding authority?